

## Assignment - 2

①

$$m \ddot{x} + k \dot{x} |\dot{x}| = u$$

(a) state space model.

$$\text{Let } x_1 = x, \quad x_2 = \dot{x}$$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\frac{k}{m} x_2 |x_2| + \frac{1}{m} u \end{cases}$$

(b) Conventional Sliding Mode Control.

$$s = c x_1 + x_2, \quad c > 0$$

$$\dot{s} = c \dot{x}_1 + \dot{x}_2$$

$$= c x_2 - \frac{k}{m} x_2 |x_2| + \frac{u}{m}$$

Proposed control law

$$u = -m c x_2 - K x_2 \operatorname{sign}(s)$$

$$\dot{s} = -\frac{K}{m} x_2 |x_2| - \frac{K}{m} \operatorname{sign}(s)$$

Reaching condition

$$s \dot{s} = -\frac{K}{m} x_2 |x_2| s - \frac{K}{m} |s|$$

$$\therefore \left| -\frac{K}{m} x_2 |x_2| s \right| \leq \frac{K}{m} \dot{x}_2^2 |s|$$

$$\therefore \dot{S} \leq \frac{k_2}{m} \dot{x}_2^2 |S| - \frac{K}{m} |S|$$

$$= \frac{1}{m} (k_2 \dot{x}_2^2 - K) |S|$$

$$\eta\text{-reachability} \Rightarrow \dot{S} \leq -\eta |S|, \text{ with } K \geq k_2 \dot{x}_2^2 + m\eta$$

Control law:

$$\begin{cases} u = -m c x_2 - (k_2 \dot{x}_2^2 + m\eta) \operatorname{sign}(S), \eta > 0 \end{cases}$$

(d.)

$$S = c x_1 + x_2, \quad c > 0$$

$$\dot{S} = c \dot{x}_1 + \dot{x}_2$$

$$= c x_2 - \frac{k}{m} x_2 |x_2| + \frac{u}{m} + \dot{x}_2 = \dot{x}_2$$

$$V = \frac{1}{2} S^2 > 0$$

$$\dot{V} = S \dot{S} = \frac{k}{m} x_2 |x_2| S - \frac{K}{m} |S|$$

$$(\text{using hint}) \left| \frac{k}{m} x_2 |x_2| \cdot S \right| \leq \frac{k}{m} |S| \dot{x}_2^2$$

$$\dot{V} \leq \left( \frac{k \dot{x}_2^2 - K}{m} \right) |S|$$

Hence, the sliding variable ( $S$ ) reaches zero in finite time.



$$s \geq 0 \Rightarrow x_2 = -cx_1$$

$$\dot{x}_1 = x_2 = -cx_1$$

$$x_1(t) = x_1(0) e^{-ct}, \quad c > 0$$

$$x_1(t) \rightarrow 0 \text{ as } t \rightarrow \infty$$

and

$$x_2 = -cx_1$$

$$= -cx_1(t) e^{-ct}, \quad c > 0$$

$$x_2(t) \rightarrow 0 \text{ as } t \rightarrow \infty$$

so, states  $x_1, x_2$  reaches zero asymptotically

$$0 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

3.

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 3 & 5 \end{bmatrix} x + \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} (u+d)$$

$$x \in \mathbb{R}^3, u \in \mathbb{R}^2, d \in \mathbb{R}^2, d = \begin{bmatrix} 2 \sin(t) + \cos(t) \\ 1 + \cos(t) \end{bmatrix}$$

(A) Equilibrium point for  $d=0$

at equilibrium  $\dot{x} = 0$

$$Ax + Bu = 0$$

for  $u=0$

$$Ax = 0$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 3 & 5 \end{bmatrix} x = 0$$

$$x_2 = 0$$

$$x_3 = 0$$

$$x_1 + 3x_2 + 5x_3 = 0 \Rightarrow x_1 = 0$$

$$\left\{ x_{eq} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, u = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right.$$



# Continuous State Feedback Controller

$$u = -Kx.$$

check controllability

$$C = [B \quad AB \quad A^2B]$$

$$= \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 0 & 1 & 2 \\ 1 & 2 & 10 & 11 \end{bmatrix}$$

Rank = 3  $\rightarrow$  controllable

$$\begin{cases} u = -Kx. \end{cases}$$

Sliding Mode Controller

$$s = Cx, \quad C \in \mathbb{R}^{2 \times 3}$$

$$\dot{s} = C\dot{x} = C(Ax + B(u+d))$$

$$\dot{s} = CAx + CB(u+d)$$

$$\dot{s} = 0 \Rightarrow u = -(CB)^{-1} CAx.$$

for  $-K \text{ sign}(s)$

$$u = -(CB)^{-1} CAx - K \text{ sign}(s)$$

④

NTSM.

$$s = x_1 + \frac{1}{\beta} \text{sign}(x_2) |x_2|^{\frac{p}{q}}$$

$$\dot{s} = \dot{x}_1 + \frac{1}{\beta} \frac{p}{q} |x_2|^{\frac{p}{q}-1} \dot{x}_2$$

$$= x_2 + \frac{1}{\beta} \frac{p}{q} |x_2|^{\frac{p}{q}-1} [f(x) + g(x)(u+d)]$$

set  $\dot{s} = -k \text{sign}(s)$

$$\begin{cases} u = -\beta \frac{q}{p} \times \frac{1}{g} |x_2|^{2-\frac{p}{q}} \text{sign}(x_2) - \frac{f(x)}{g} - \beta \frac{q}{p} \frac{1}{g} |x_2|^{1-\frac{p}{q}} \text{sign}(s) \end{cases}$$

Let  $V = \frac{1}{2} s^2$

$$\dot{V} = s \dot{s} = s(-k \text{sign}(s) + d)$$

$$= -k |s| + d s \leq -(k - g d_{\max}) |s|$$

with  $k > g d_{\max} + \eta$ .

$$\dot{V} \leq -\eta |s| = -\eta \sqrt{2V} \rightarrow \text{finite time stable}$$

convergence time.

After  $\delta = 0$

$$\dot{x}_2 = -\frac{1}{p} \beta |x_2|^{2-\frac{p}{q}} \text{sign}(x_2).$$

let  $\beta = 1$ ,  $p = 5$ ,  $q = 3$ .

$$\dot{x}_2 = -\frac{3}{5} |x_2|^{\frac{1}{3}} \text{sign}(x_2)$$

$$\left. \frac{3}{2} x_2^{\frac{2}{3}} \right|_{x_2}^0 = \frac{3}{5} T$$

$$T = \frac{5}{2} |x_2(t)|^{\frac{2}{3}}.$$

for  $t_2 \rightarrow |s(t)| \leq |s(0)| - \eta t$ .

$$t = \frac{|s(0)|}{\eta}$$

$$\left\{ \begin{array}{l} T_{\text{total}} = \frac{|s(0)|}{\eta} + \frac{5}{2} |x_2(t)|^{\frac{2}{3}} \\ , \end{array} \right.$$